
DEPARTMENT OF MATHEMATICS

Mathematics Lab Assessment No. 2

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Question	An automated assembly line uses robotic grippers to assemble electronic control units. Historical maintenance records indicate that gripper failures occasionally occur due to wear, contamination, or pneumatic leakage. The maintenance department has collected the following information: • Probability that a gripper fails during a production shift = 0.04 • If a gripper is failing, the monitoring system generates a warning signal with probability 0.92 • If a gripper is operating normally, the monitoring system still generates a false warning with probability 0.05 Calculate: 1. Probability that a gripper is operating normally. 2. Probability that the monitoring system generates a warning. 3. Probability that a gripper is actually failing given a warning. 4. Probability of a false warning. 5. Reliability of the monitoring system.
Solution	i) Identify the parameters and mathematical concept $P(F)$ = Probability of failure = 0.04 $P(N)$ = Probability of normal operation = 0.96 $P(W/F)$ = Probability of warning given failure = 0.92 $P(W/N)$ = Probability of false warning = 0.05 Mathematical Concepts • Law of Total Probability • Bayes' Theorem ii) Solve analytically

Lab-2.

Given

$$P(F) = 0.04$$

$$P(W/F) = 0.92$$

$$P(W/N) = 0.05$$

i.e. probability that gripper is operating normally is

$$P(N) = 1 - P(F)$$

$$= 1 - 0.04$$

$$P(N) = 0.96$$

$\Rightarrow$ 96% of production shift occurs without gripper failure

ii.e. Probability that the monitoring system generates a warning

$$P(W) = P(W/F)P(F) + P(W/N)P(N)$$

$$P(W) = (0.92)(0.04) + (0.05)(0.96)$$

$$P(W) = 0.0368 + 0.048$$

$$P(W) = 0.0848$$

$\Rightarrow$ 8.48% of production shift generates a warning

(3) Probability that the gripper is actually failing given warning

$$P(F/W) = \frac{P(W/F)P(F)}{P(W)}$$

$$P(F/W) = \frac{0.92 \times 0.04}{0.0848}$$

$$P(F/W) = 0.43396$$

Only about 43.39% of warning signals corresponding to actual failure.

ii.e. probability of a false warning

$$\begin{aligned} P(\text{False warning}) &= P(W/N)P(N) \\ &= (0.05)(0.96) \\ &= 0.048 \\ &= 4.8\% \end{aligned}$$

iii.e. reliability of the monitoring system reliability

$$\begin{aligned} &= P(F/W) \cdot P(N) \\ &= 43.39\% \end{aligned}$$

This means less than half of all warning indicate a true failure

iii) GeoGebra Screenshot / Program Execution

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# Given Data
P_F = 0.04      # Probability of failure
P_W_F = 0.92    # Warning given failure
P_W_N = 0.05    # False warning probability

# Step 1: Probability of normal operation
P_N = 1 - P_F

# Step 2: Probability of warning
P_W = (P_W_F * P_F) + (P_W_N * P_N)

# Step 3: Bayes Theorem
P_F_W = (P_W_F * P_F) / P_W

# Step 4: False warning probability
P_false_warning = P_W_N * P_N

# Step 5: Reliability of monitoring system
reliability = P_F_W * 100

# Display Results
print("Probability of Normal Operation P(N):", round(P_N, 4))
print("Probability of Warning P(W):", round(P_W, 4))
print("Probability of Failure Given Warning P(F|W):", round(P_F_W, 4))
print("Probability of False Warning:", round(P_false_warning, 4))
print("Monitoring System Reliability:", round(reliability, 2), "%")
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iv) Results and analysis from the graph

Probability of Normal Operation $P(N)$: 0.96
Probability of Warning $P(W)$: 0.0848
Probability of Failure Given Warning $P(F|W)$: 0.434
Probability of False Warning: 0.048
Monitoring System Reliability: 43.4 %

- The monitoring system has a reliability of 43.39%, indicating that a warning signal corresponds to an actual gripper failure only about 43.39% of the time.
- Because fewer than half of the warning signals represent genuine failures, performing maintenance immediately after every warning may result in unnecessary inspections and increased maintenance costs.
- Even though the monitoring system has a high detection rate (92%), the actual failure occurrence is relatively low (4%), and the presence of a 5% false alarm rate causes many warnings to be triggered when the gripper is functioning normally.

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	• When failure events are infrequent, the number of false alarms becomes comparatively higher, which lowers the confidence that a warning truly indicates a fault. • As the frequency of actual failures increases, the likelihood that a warning signal reflects a real problem also increases, thereby improving the effectiveness and credibility of the monitoring system. • Engineers should use warning signals as an initial indicator rather than the sole basis for maintenance decisions. To make informed decisions, warning signals should be supported by: • Additional diagnostic tests • Previous maintenance records • Analysis of sensor data trends • Physical inspection of the equipment • Other relevant condition-monitoring information before scheduling maintenance activities